Introductory Seminar 2026

Operations Research Case

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Battery Optimisation

Contact

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Operations Research

What is OR?

Operations Research (OR) is a scientific discipline that deals with the application of advanced analytical methods to help make better decisions.

OR Techniques

Many problems can be solved by applying OR techniques, such as:

- Linear Programming
- Combinatorial Optimization
- Simulation

The Setting

System Components

Solar panels: Generate electricity during the day and may exceed household demand.

Home battery: Stores surplus solar energy or cheap grid electricity for later use.

Dynamic contract: Electricity prices vary hourly.

Net demand: Consumption minus solar production (negative = surplus, positive = grid purchase).

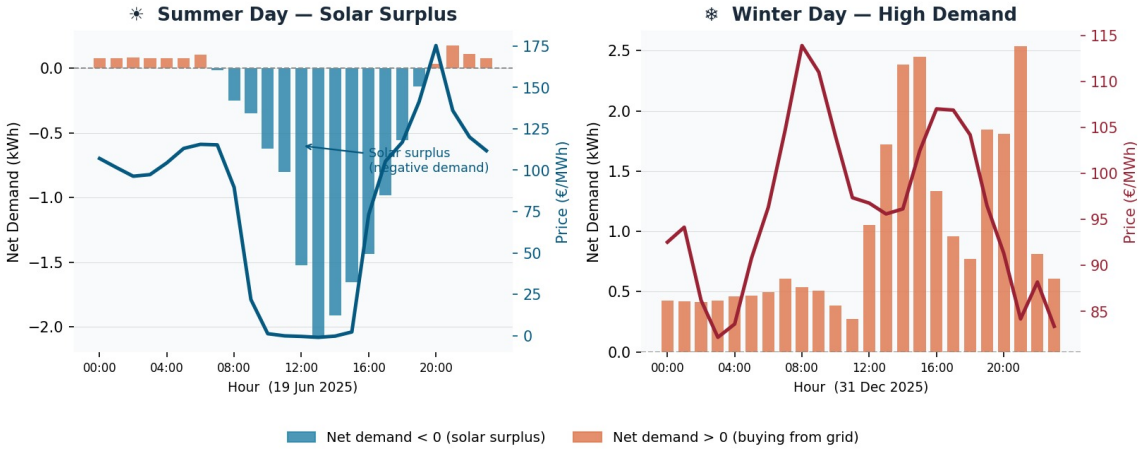
The Decision Problem

Determine the optimal charging and discharging strategy over time.

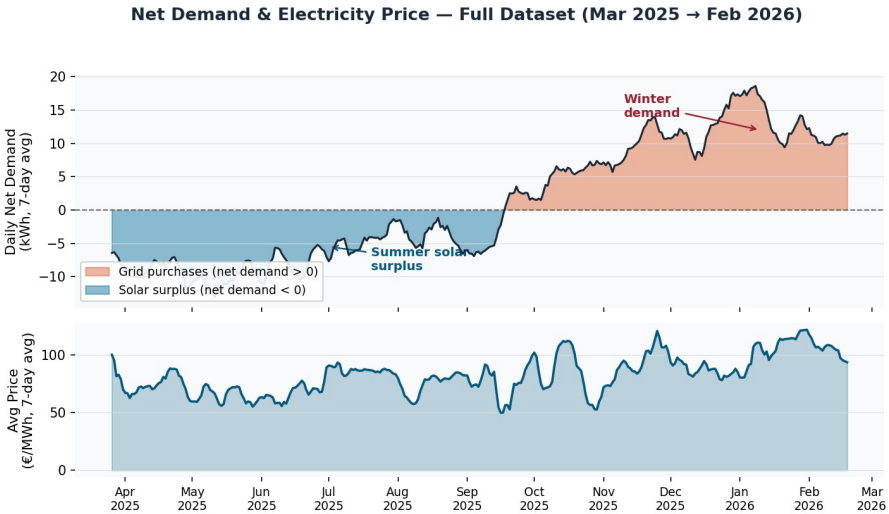
Exploit price differences by charging when prices are low and discharging when prices are high.

Objective: Maximise total household profit.

Summer vs. Winter



A Year in Data



The Optimization Problem

Objective

Maximise total household profit over all time periods by determining the optimal charging and discharging decisions.

Key Technical Constraints

(1) State-of-Charge Limits: The battery level must remain within minimum and maximum bounds.

(2) Charging & Discharging Power Limits: Charging and discharging rates are capped.

(3) Efficiency Losses: Energy losses occur during charging and discharging.

(4) No Simultaneous Charging & Discharging: A binary variable enforces mutually exclusive operation.

Two MIP Formulations

Goal: Compare two formulations of the same battery optimization problem.

Formulation 1 — Basic Operation MIP

Elgersma et al. (2024), §II.A
Standard formulation using binary variables. Straightforward and intuitive.

Formulation 2 — Tighter Operation MIP

Elgersma et al. (2024), §III.B
Alternative formulation with tighter LP relaxation bounds.

Your task: Implement both models and explain the computational differences.

Research Proposal

Submit before **March 26, 11:59**

Formulate the problem with 2 different MIPs. Report your ideas by:

- Defining decision variables
- Defining sets and parameters
- Formulating the objective function (with explanation)
- Formulating the constraints (with explanation)

Submission details

- Hand in **at most 2 pages**
- Discussed at the first compulsory meeting on **Friday, March 27**

Progress Report

Submit before **April 1, 11:59**

Report your progress on **all assignments** in the case

Present ideas for extensions (if you have any)

Formulate remaining questions (if you have any)

Submission details

Hand in **at most 1 page**

Discussed at the second compulsory meeting on **Thursday, April 2**

Deadline — Final Report

1st Version

April 10, 11:59

Peer Review

Deadline for all groups: April 15, 11:59

- **FEB22009 / FEB22009X:** during Academic Skills
- **FEB22009Q:** via Canvas
- **FEB22009S:** via e-mail

2nd Version (Final)

April 17, 23:59